



Effect of oculo-motor and gaze stability exercises on postural stability and dynamic visual acuity in healthy young adults

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ABSTRACT

The purpose of this study was to investigate the effectiveness of oculo-motor exercises and gaze stability exercises on postural stability and dynamic visual acuity (DVA) in healthy young adults. Subjects were randomly divided into experimental ($n = 28$) and control groups ($n = 13$). The experimental group performed oculo-motor exercises and gaze stability exercise for three weeks and the control group did not. Pre and post postural stability during quiet standing, standing with active head rotation, and DVA were measured on both groups. Significant differences were found in postural stability during standing with active head rotation and DVA in the experimental group ($p < 0.05$). No significant differences were found in postural stability or DVA in the control group. Improvements in postural stability and DVA after three weeks of oculo-motor exercises and gaze stability exercises suggest that this specific type of vestibular and ocular system exercises may be beneficial for healthy young adults.

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1. Introduction

The vestibular system contributes to postural stability and visual stabilization through the vestibulo-spinal reflex (VSR) and vestibulo-ocular reflex (VOR), respectively [1]. The VSR produces contraction of anti-gravity muscles to help maintain postural stability and the VOR produces eye movements to maintain clear visual acuity during head movements [1]. Additionally, the cervico-ocular reflex (COR) interacts with the vestibular nuclei and can also produce eye movements [2]. The VSR and VOR are activated during stimulation of the vestibular organs while the COR is activated during stimulation of the upper neck joint receptors [3]. As a result, both head and neck rotation contribute to activation of the VSR, VOR, and COR [1,2]. Decreased postural stability and visual acuity can be induced by active neck rotation and/or head tilt in patients with vestibular dysfunction as well as in healthy people [4–11]. The effects of vestibular rehabilitation gaze stability exercises in patients with vestibular dysfunction is well established and includes improved postural stability and dynamic visual acuity (DVA), however there are no reports that gaze stability exercises are beneficial for healthy populations [12–17].

The purpose of this study was to investigate the effects of gaze stability exercises on static and dynamic postural stability and DVA in healthy young adults.

2. Materials and methods

2.1. Subjects

Forty-one healthy young adults were recruited for this study and randomly divided using a random number table into experimental ($n = 28$, ages 19.1 ± 0.6 years) and control groups ($n = 13$, mean age 19.4 ± 1.2 years). The experimental group performed gaze stability exercise for three weeks and the control group did not. Informed consent was obtained in compliance with Nihon Fukushi University Review Board protocol prior to participation in the study.

2.2. Intervention

We used oculo-motor exercises and gaze stability exercises previously described by Herdman [1] and Cawthorne and Cooksey [18,19]. A total of four different exercises were performed including (1) saccadic eye movement exercises; (2) smooth pursuit exercises; (3) adaptation X1 exercises; and (4) adaptation X2 exercises. The saccadic eye movement exercise included moving the eyes horizontally between two stationary targets while keeping the head still. The smooth pursuit exercise included moving the target horizontally and tracking it with the eyes while keeping the head still. The adaptation X1 exercise included moving the head horizontally while keeping the stationary target in focus. The adaptation X2 exercise included moving the head and target in opposite directions horizontally while tracking the target with the eyes.

During the exercises, subjects were instructed to focus on the target (a letter on the students' ID card) held in their hand and to move the target or their head as fast as they could while maintaining clear focus on the target. Subjects performed each

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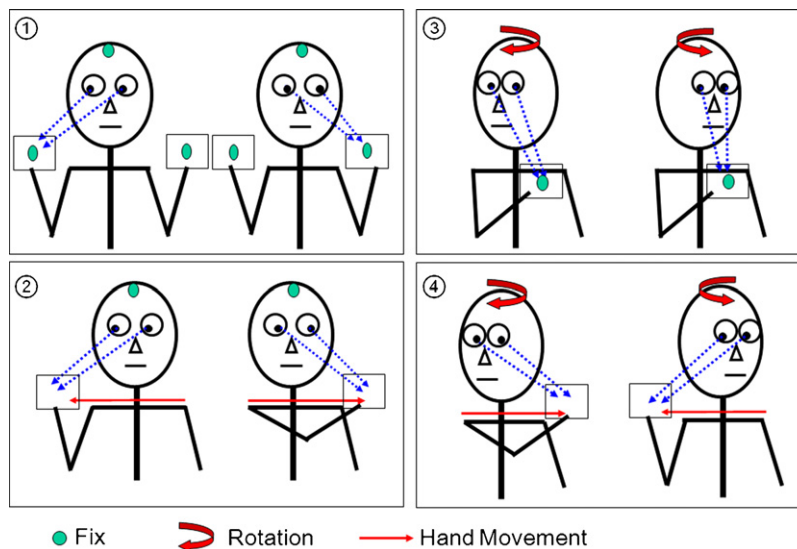


Fig. 1. Oculo-motor exercises and gaze stability exercise illustrations and descriptions. The target was held in the subjects' hand approximately thirty centimeters away from the subjects' eyes. Subjects performed all exercise in a sitting position for 5 min daily for three weeks. Subjects were instructed to move the target, or their head, as fast as they could while maintaining clear focus on the target during the following exercises: saccadic eye movements, smooth pursuit, adaptation X1, and adaptation X2.



Fig. 2. Postural stability examination. Subjects stood on a CDP force plate and fixed their eyes on a light spot (0.7 cm diameter) that was projected onto a screen in front of the subject at eye level.

exercise for 5 min daily for three weeks in the experimental group, and all exercises were performed in a seated position in a classroom together. See Fig. 1.

2.3. Postural stability measurement

Postural stability was measured using a computerized dynamic posturography (CDP) system (Balance Master[®], NeuroCom[®], Clackamas, Oregon). Subjects stood on the CDP force plate in a dark room to eliminate the visual stimulus, and fixed their eyes on a light spot (0.7 cm diameter) that was projected onto a screen in front of the subject at eye level (Fig. 2). Static (quiet standing) and dynamic (standing with active head rotation) conditions were measured. The static condition was measured with subjects standing on the CDP force plate quietly. The dynamic condition was measured with subjects standing with active neck rotation that was standing on the CDP force plate while rotating their neck in the yaw plane at a frequency of 2 Hz. We used a metronome to keep the neck rotation frequency at 2 Hz and neck rotation range was a total of 70° during postural stability testing. Before testing the postural stability, subjects practiced three times to remember the neck rotation range. Three trials were taken for each condition and a calculated mean velocity of anterior-posterior and medio-lateral body sway was determined. Each recording took 10 s, followed by a ten second rest between trials, and a 30 s rest between the two different conditions.

2.4. Dynamic visual acuity (DVA) measurement

DVA was measured using the technique described by Herdman et al. [6] and Hillman et al. [11]. DVA measurement was reading each number on the paper during active neck rotation.

Using Microsoft PowerPoint, we created 10 pieces of paper that each had 5 random numerals, ranging from 12 to 20 point font, printed on the center of the page. Each paper had a total of five numerals at one font size and subjects were assigned ten pieces of paper. Subjects sat in a chair with the back rest 70 cm away from the paper which was positioned at eye level. Subjects were asked to read each number during active neck yaw rotation at a frequency of 2 Hz (Fig. 3). We used a metronome and gave subjects verbal feedback to keep the frequency at approximately 2 Hz during the testing. The DVA percent correct score was calculated by totaling the number of answers. Subjects were given 5 s to answer during active neck yaw rotation for each paper. Before testing DVA we confirmed that all subjects were able to read the smallest font size of 12 from 70cm without head movements.

2.5. Statistical analysis

Statistical analysis was performed using the Mann Whitney *U*-test and Student's *t*-test for determining baseline differences between groups for age, gender, pre postural stability and pre DVA, and a paired *t*-test for determining pre-post

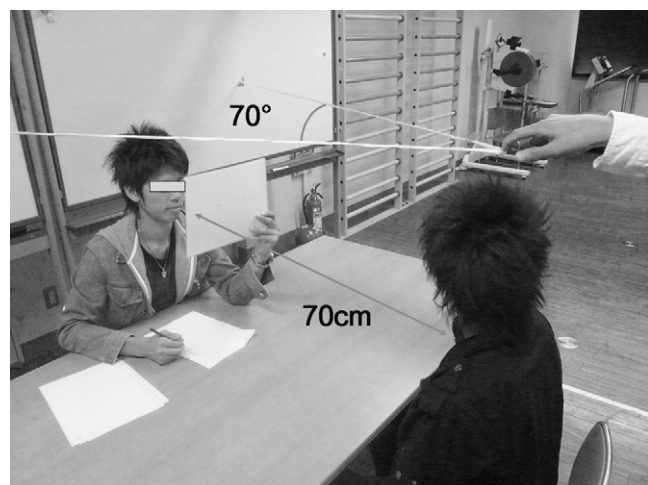


Fig. 3. Dynamic visual acuity examination. Subjects sat in a chair with back rest 70 cm away from the paper and read the numbers during neck rotation (for example: 30572). Total neck rotation range of motion was 70° and the frequency was approximately 2 Hz (paced by metronome).

differences of postural stability and DVA between groups. A level of significance for all analysis was set at $p < 0.05$.

3. Results

There was no significant difference between groups in age, gender, pre postural stability, or pre-dynamic visual acuity (Table 1). In the experimental group, post gaze stability exercise training analysis revealed significant differences in postural stability during standing with active neck rotation (Table 2) and DVA (Table 2). In the control group, no significant differences were found in postural stability or DVA (Table 2).

4. Discussion

There are many reported benefits of gaze stability exercises for patients with vestibular dysfunction [12–17]. Strupp et al. [17] reported significant reduction of body sway caused by improved

VSR compensation in patients with vestibular dysfunction as a result of specific vestibular rehabilitation exercises. However, there are no reports on the effects of vestibular rehabilitation or oculo-motor exercises and gaze stability exercises in healthy young adults. Previous investigations have demonstrated that active head movements decrease postural control and postural stability in patients with vestibular dysfunction as well as in healthy subjects [4,5].

Head movements stimulate the VOR which generates eyes movements of the same speed but opposite direction for maintenance of visual acuity [1]. Rapid head movements can cause retinal slip, and when retinal slip exceeds 2–4°/s, degradation of visual acuity occurs [6–10]. Herdman et al. [7] reported a significant difference between DVA-predictable and DVA-unpredictable movements. DVA-predictable was better than DVA-unpredictable and the difference was theorized to be the result of central preprogramming. Moreover, Collewyn et al. [20] and Miles and Eighmy [21] suggested that visual and vestibular stimulation could improve the function of vestibular system due to a neural adaptation at the vestibular nuclei [22] and cerebellum [23]. Therefore, central pre-programming and neural adaptation may influence the improvement of DVA through oculo-motor exercises and gaze stability exercises.

Improvements in postural stability with active head rotation and DVA after three weeks of oculo-motor exercises and gaze stability exercises were observed in healthy young adults. We speculate that these improvements were the result of improved vestibulo-spinal function and neural adaptation of the vestibular nuclear complex and/or cerebellum, as well as enhanced central pre-programming. We theorize that quiet standing was not improved in the experimental group due because the gaze stability exercise program did not incorporate quiet standing activities. All exercises involved head and/or eye movements. Perhaps if we had included a specific quiet standing exercise, significant improvements would have been observed.

Limitations of the study included that we did not monitor the exact head speed and frequency during neck rotation during the examination of postural stability and DVA. Future investigations should consider using an accelerometer to monitor and establish exact head movements speed and frequency during neck rotation. Also, we did not have control group who performed any specific

Table 1
Comparison of baseline data between groups.

	Experimental group	Control group	<i>p</i> -value
Age (year)	19.1 ± 0.6	19.4 ± 1.2	0.531 ^a
Gender	Male:13 female:15	Male:5 female:8	0.075 ^a
Quiet standing (°/s)	0.25 ± 0.01	0.26 ± 0.12	0.680 ^b
Standing with active neck rotation (°/s)	1.11 ± 0.51	0.93 ± 0.45	0.188 ^b
DVA (% correct)	41.8 ± 35.1	59.3 ± 29.1	0.133 ^b

Values are mean ± SD

Quiet standing: standing on the CDP force plate quietly. Standing with active neck rotation: standing on the CDP force plate while rotating their neck at a frequency of 2 Hz. DVA: reading each number on the paper during active neck rotation at a frequency of 2 Hz in sitting.

^a Mann Whitney *U*-test.

^b Student's *t*-test.

Table 2
Comparison of postural stability and dynamic visual acuity before and after exercise program.

	Experimental group			Control group		
	Pre	Post	<i>p</i> -value	Pre	Post	<i>p</i> -value
Quiet standing (°/s)	0.25 ± 0.01	0.26 ± 0.01	0.855 ^a	0.26 ± 0.12	0.29 ± 0.11	0.827 ^a
Standing with active neck rotation (°/s)	1.11 ± 0.51	0.71 ± 0.27	0.001 ^a	0.93 ± 0.45	1.04 ± 0.32	0.557 ^a
DVA (% correct)	42.77 ± 36.22	78.5 ± 24.1	0.002 ^a	59.3 ± 29.1	53.2 ± 27.3	0.494 ^a

Values are mean ± SD

Quiet standing: standing on the CDP force plate quietly. Standing with active neck rotation: Standing on the CDP force plate while rotating their neck at a frequency of 2 Hz. DVA: reading each number on the paper during active neck rotation at a frequency of 2 Hz in sitting.

^a Paired *t*-test.

activities as a non-visual exercises, thus the improvements of postural stability and DVA might have been affected by simple head movements in daily activities. Future investigations should consider having the control group performs head movements with eyes closed. Finally, a quiet standing exercise component could be added in future investigations.

In conclusion, the results of this study showed improvements in postural stability with active head rotation and DVA after performing oculo-motor exercises and gaze stability exercises for three weeks.

Conflict of interest statement

None of the contributing authors have any conflict of interest that may have biased this work.

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